

CLINICAL PROGRAMMING BOOTCAMP • MODULE 14

From Tabulation to Analysis

Why ADaM exists — and building ADSL

You have spent ten days making data faithful. Now you make it answerable.

Hands-on: Notebook 14 (SAS) — Build ADSL



By the end of this module you can...

1

Say why ADaM exists in one sentence, without using the word 'analysis-ready' as if it explained itself.

2

Name the three dataset classes ADSL, BDS and OCCDS — and say which to reach for.

3

Build ADSL end to end treatment variables, population flags, disposition, baseline measurements.

4

Apply the ADaM baseline rule and say why copying VSBLFL breaks silently.

5

Defend traceability explain how a regulator gets from a number in a table back to a CRF page.

The same subject, told two different ways

Subject ABC-01-01-004 wants to appear in a table of mean treatment duration by arm.

SDTM asks: what happened?

DM → RFXSTDTC, RFXENDTC (text)

DS → disposition, why they left

EX → what was actually given

Three domains. Nothing derived.

*To get the duration you must join
them and compute it yourself —
every time, in every program.*

ADaM asks: what do I report?

ADSL → one row, everything on it

TRT01P = Placebo

TRTSDT = 06MAR2024

TRTEDT = 25MAR2024

TRTDURD = 20

SAFFL = Y

One dataset. PROC MEANS. Done.

Same subject. Same truth. Different organising principle — because they have different readers.

THE TEST A DATASET MUST PASS

**A table, listing or figure can be produced
by subsetting and summarising ONE dataset —
with no further merging and no further derivation.**

If you need a join at the analysis step, the analysis dataset was built wrong.

That is not a style preference. It is the difference between a result a reviewer can reproduce and one they cannot.

Three dataset classes — and how to choose

ADSL

Subject-Level

ONE row per subject

The spine. Treatment, populations, key dates.
Every other dataset copies from it.

BDS

Basic Data Structure

**One row per subject
per parameter per visit**

Anything measured repeatedly:
vitals, labs, scores. ADVS, ADLB.

OCCDS

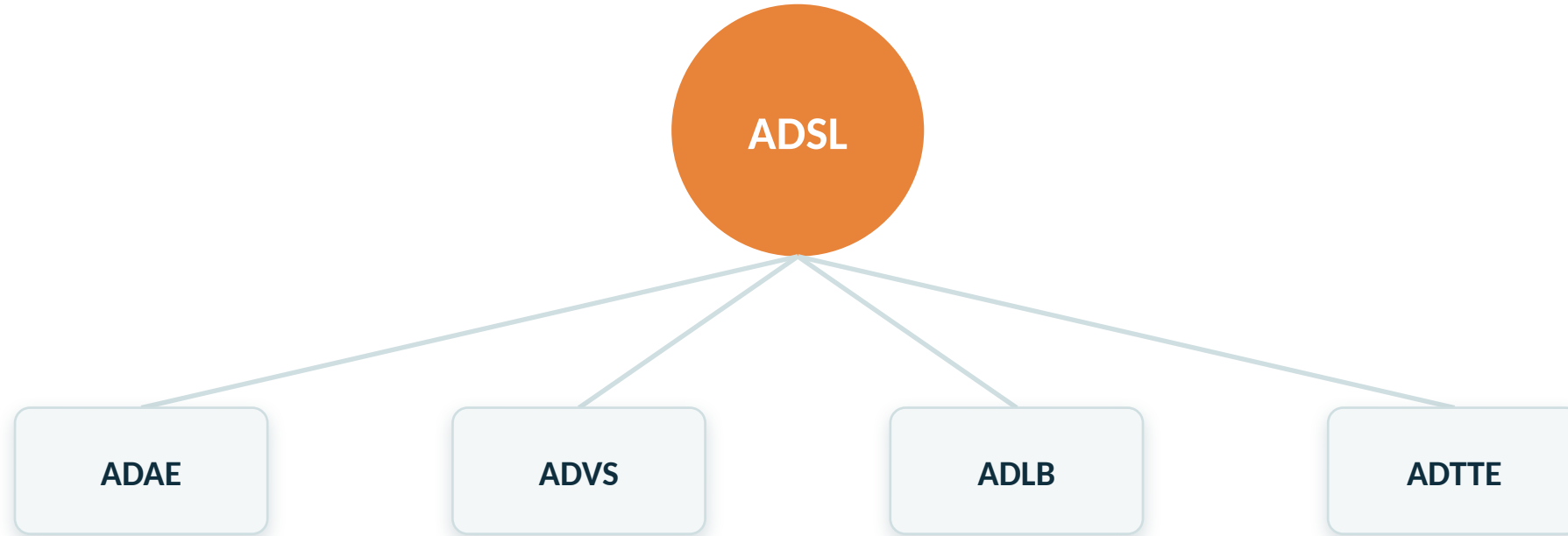
Occurrence

One row per event

Things that either happened or did not:
adverse events, medications. ADAE.

Pick by SHAPE, not by subject matter. A questionnaire score is BDS. A protocol deviation is OCCDS.

One dataset defines the study's subject-level truth



TRT01P · TRT01A · SAFFL · ITTFL · TRTSDT · TRTEDT · AGEGR1

These are COPIED downstream — never re-derived. Re-deriving invites the two to drift, and nothing tells you when they do.

In ADaM, a date is a number

SDTM

```
RFXSTDTC = "2024-03-01"
```

```
CHARACTER. ISO 8601.
```

```
Faithful, readable, sortable.
```

```
You cannot subtract it.
```

ADaM

```
TRTSDT = 23436  
        (01MAR2024)
```

```
NUMERIC SAS date.
```

```
TRTEDT - TRTSDT + 1 = 28
```

INPUT () converts character → numeric. **PUT ()** converts numeric → character.

Getting these backwards is the most common SAS error in this course. It does not always fail loudly.

Why the change? Because a table needs arithmetic, and arithmetic needs numbers.

Baseline is a question about time, not about a visit

Baseline = the LAST non-missing value on or before the date of first dose.

The tempting shortcut

```
if vsblfl = 'Y';
```

HEIGHT is collected only at SCREENING.
VSBLFL marks the BASELINE visit.

Zero height records carry VSBLFL='Y'.
All 8 subjects lose baseline height —
and their BMI. No error. No warning.

The rule, done properly

```
where vsdte <= trtsdt  
... if last.vstestcd;
```

HEIGHT → resolves to Screening
WEIGHT → resolves to Baseline

Same rule. Different visits.
Because it asks about TIME.

A rule written in terms of visit labels breaks the moment a parameter is collected on a different schedule.

Different questions that happen to have the same answer

SAFFL

Safety

Did the subject take at least one dose?

`TRTSDT is not missing`

Y for all 8

ITTFL

Intent-to-treat

Was the subject randomised?

`RANDDT is not missing`

Y for all 8

COMPLFL

Completers

Did the subject finish the study?

`DS disposition = COMPLETED`

Y for all 8

All three are 'Y' for every ABC-01 subject — and all three are still derived from different rules.

A screen failure, or a subject randomised but never dosed, separates them. Then the tables built on each separate too.

A regulator must get from a table back to a CRF page



Every link must be documented, and every derivation must be reproducible from the link below it.

This is why ADaM carries SRCDOM / SRCVAR / SRCSEQ. A record that names its source SDTM row can be checked. A derived record with no single source leaves them blank — and that blank is information, not an omission.

Six ADSL mistakes, and what each one costs

Copying VSBLFL as baseline

Every subject loses baseline height and BMI. No error.

Re-deriving SAFFL downstream

Two answers to 'who was dosed'. Neither defensible.

DCSREAS filled for completers

'Discontinued for reason X' becomes uncountable.

TRTDURD without the +1

Every duration off by one; a one-day subject reads as zero.

Duplicate USUBJID

Silently multiplies rows in every downstream merge.

Dates left as character

Arithmetic impossible; study day cannot be derived.

Five of these six produce no error message. That is the recurring lesson of this course.

WHAT'S NEXT

Build the spine first

1

Step 1 • Notebook 14 Build ADSL from DM, DS and VS. Eight rows, and one of them discontinued.

2

Step 2 • Exercise 1 Do the baseline the WRONG way on purpose. Watch it fail silently.

3

Step 3 • Check PROC COMPARE against data/adam/adsl.csv — the finished reference.

Everything in the next four days copies from what you build today. Get ADSL right and the rest is bookkeeping.